Musterlösung - Wellengleichung B-Feld

Maxwell-Gleichungen ohne Quellen

$$\operatorname{div}(\vec{E}) = \cancel{\rho(\vec{r})/\varepsilon_0} \qquad (1)$$

$$\operatorname{rot}(\vec{E}) = -\partial_t \vec{B} \qquad (2)$$

$$\operatorname{div}(\vec{B}) = 0 \qquad (3)$$

$$\operatorname{rot}(\vec{B}) = \cancel{\mu_0 \vec{j}} + \mu_0 \varepsilon_0 \partial_t \vec{E} \qquad (4)$$

Herleitung Wellengleichung für das B-Feld

$$\partial_t^2 \vec{B} = -\partial_t \operatorname{rot}(\vec{E}) = -\operatorname{rot}(\partial_t \vec{E}) = -\operatorname{rot}\left(\frac{1}{\mu_0 \varepsilon_0} \operatorname{rot}(\vec{B})\right)$$

$$= \frac{-1}{\mu_0 \varepsilon_0} \operatorname{rot}(\operatorname{rot}(\vec{B})) = \frac{-1}{\mu_0 \varepsilon_0} \left(\operatorname{grad}(\underbrace{\operatorname{div}(\vec{B})}_{=0}) - \vec{\nabla}^2 \vec{B}\right)$$

$$= \frac{1}{\mu_0 \varepsilon_0} \vec{\nabla}^2 \vec{B}$$

$$\hookrightarrow \quad \boxed{\frac{1}{c^2} \partial_t^2 \vec{B} - \vec{\nabla}^2 \vec{B} = 0} \quad \text{mit} \quad c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

Herleitung von $\operatorname{grad}(\operatorname{div}(\vec{A})) - \Delta\vec{A} = \operatorname{rot}(\operatorname{rot}(\vec{A}))$

Linke Seite

$$\vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \Delta\vec{A} = \begin{pmatrix}\partial_x\\ \partial_y\\ \partial_z\end{pmatrix}(\partial_x A_x + \partial_y A_y + \partial_z A_z) - \begin{pmatrix}(\partial_x^2+\partial_y^2+\partial_z^2)A_x\\ (\partial_x^2+\partial_y^2+\partial_z^2)A_y\\ (\partial_x^2+\partial_y^2+\partial_z^2)A_z\end{pmatrix}$$

$$= \begin{pmatrix}\partial_x^2 A_x + \partial_x\partial_y A_y + \partial_x\partial_z A_z\\ \partial_y\partial_x A_x + \partial_y^2 A_y + \partial_y\partial_z A_z\\ \partial_z\partial_x A_x + \partial_z\partial_y A_y + \partial_z^2 A_z\end{pmatrix} - \begin{pmatrix}(\partial_x^2+\partial_y^2+\partial_z^2)A_x\\ (\partial_x^2+\partial_y^2+\partial_z^2)A_y\\ (\partial_x^2+\partial_y^2+\partial_z^2)A_z\end{pmatrix}$$

$$= \begin{pmatrix}\partial_x\partial_y A_y + \partial_x\partial_z A_z - \partial_y^2 A_x - \partial_z^2 A_x\\ \partial_y\partial_x A_x + \partial_y\partial_z A_z - \partial_x^2 A_y - \partial_z^2 A_y\\ \partial_z\partial_x A_x + \partial_z\partial_y A_y - \partial_x^2 A_z - \partial_y^2 A_z\end{pmatrix}$$

identisch ✓

Rechte Seite

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \begin{pmatrix}\partial_x\\ \partial_y\\ \partial_z\end{pmatrix}\times\left(\begin{pmatrix}\partial_x\\ \partial_y\\ \partial_z\end{pmatrix}\times\begin{pmatrix}A_x\\ A_y\\ A_z\end{pmatrix}\right) = \begin{pmatrix}\partial_x\\ \partial_y\\ \partial_z\end{pmatrix}\times\begin{pmatrix}\partial_y A_z - \partial_z A_y\\ \partial_z A_x - \partial_x A_z\\ \partial_x A_y - \partial_y A_x\end{pmatrix}$$

$$= \begin{pmatrix}\partial_y(\partial_x A_y - \partial_y A_x) - \partial_z(\partial_z A_x - \partial_x A_z)\\ \partial_z(\partial_y A_z - \partial_z A_y) - \partial_x(\partial_x A_y - \partial_y A_x)\\ \partial_x(\partial_z A_x - \partial_x A_z) - \partial_y(\partial_y A_z - \partial_z A_y)\end{pmatrix}$$

$$= \begin{pmatrix}\partial_y\partial_x A_y - \partial_y^2 A_x - \partial_z^2 A_x + \partial_z\partial_x A_z\\ \partial_z\partial_y A_z - \partial_z^2 A_y - \partial_x^2 A_y + \partial_x\partial_y A_x\\ \partial_x\partial_z A_x - \partial_x^2 A_z - \partial_y^2 A_z + \partial_y\partial_z A_y\end{pmatrix}$$